

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457740.67056 +/- 0.00035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.045
Transit depth (Rp/Rs)^2	2.6e-05 +/- 0.0045 [%]
Orbital Inclination (inc)	78.25 +/- 2.93
Airmass coefficient 1 (a1)	1.74 +/- 1.04
Airmass coefficient 2 (a2)	-0.4 +/- 0.42
Scatter in the residuals of the lightcurve fit is	60.41 %
Best Comparison Star	None
Optimal Aperture	2.41
Optimal Annulus	3.0
Transit Duration (day)	0.0259 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.045 vs 0.1178 (deviation 1.97σ)
Duration fit vs published	0.0259 d vs 0.125 d (deviation 10.61σ)
Mid-time O–C	-1572.5 min
Depth SNR	0.0
Residual scatter	60.41 %

Light curve

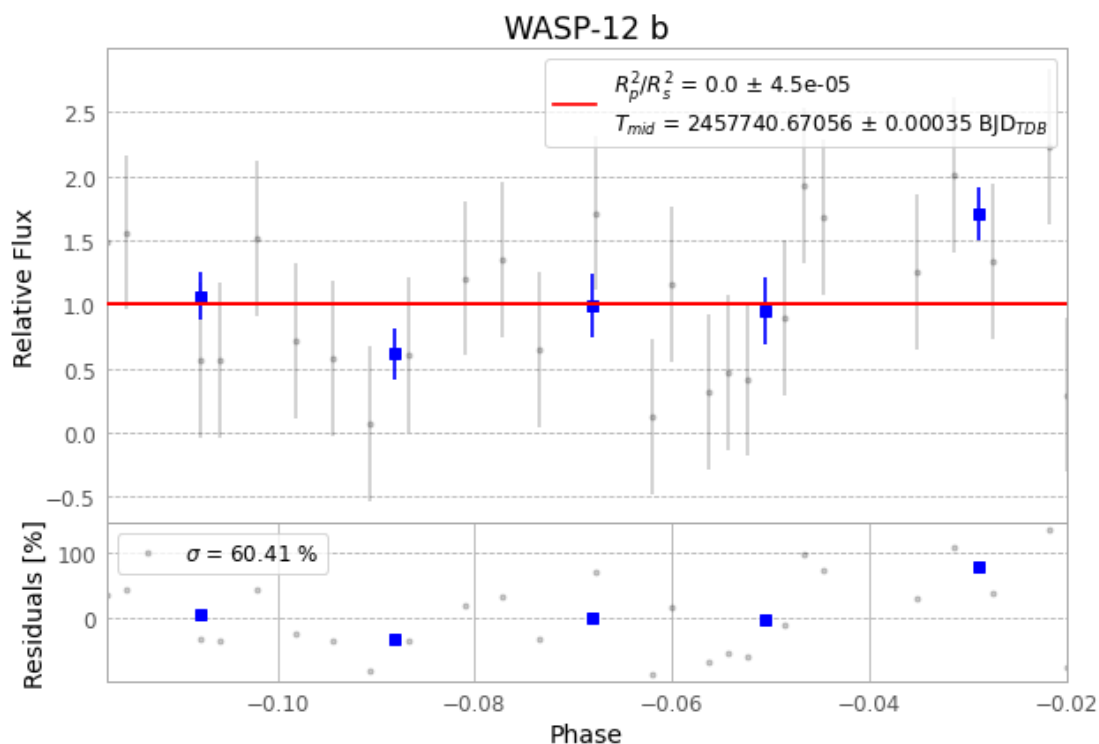


Figure 1 — FinalLightCurve_WASP-12 b_2016-12-18.png (SHA-256 9ea8b3e514eed987...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [354, 298], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 66443fa26a002443...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

61 FITS frames (40 MB), WASP-12161219025712.FITS → WASP-12161219055712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161219.log (6c2ae6d63c56...), FinalLightCurve_WASP-12 b_2016-12-18.png (9ea8b3e514ee...), FinalLightCurve_WASP-12 b_2016-12-18.csv (da64364c72b1...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 14:30Z.